

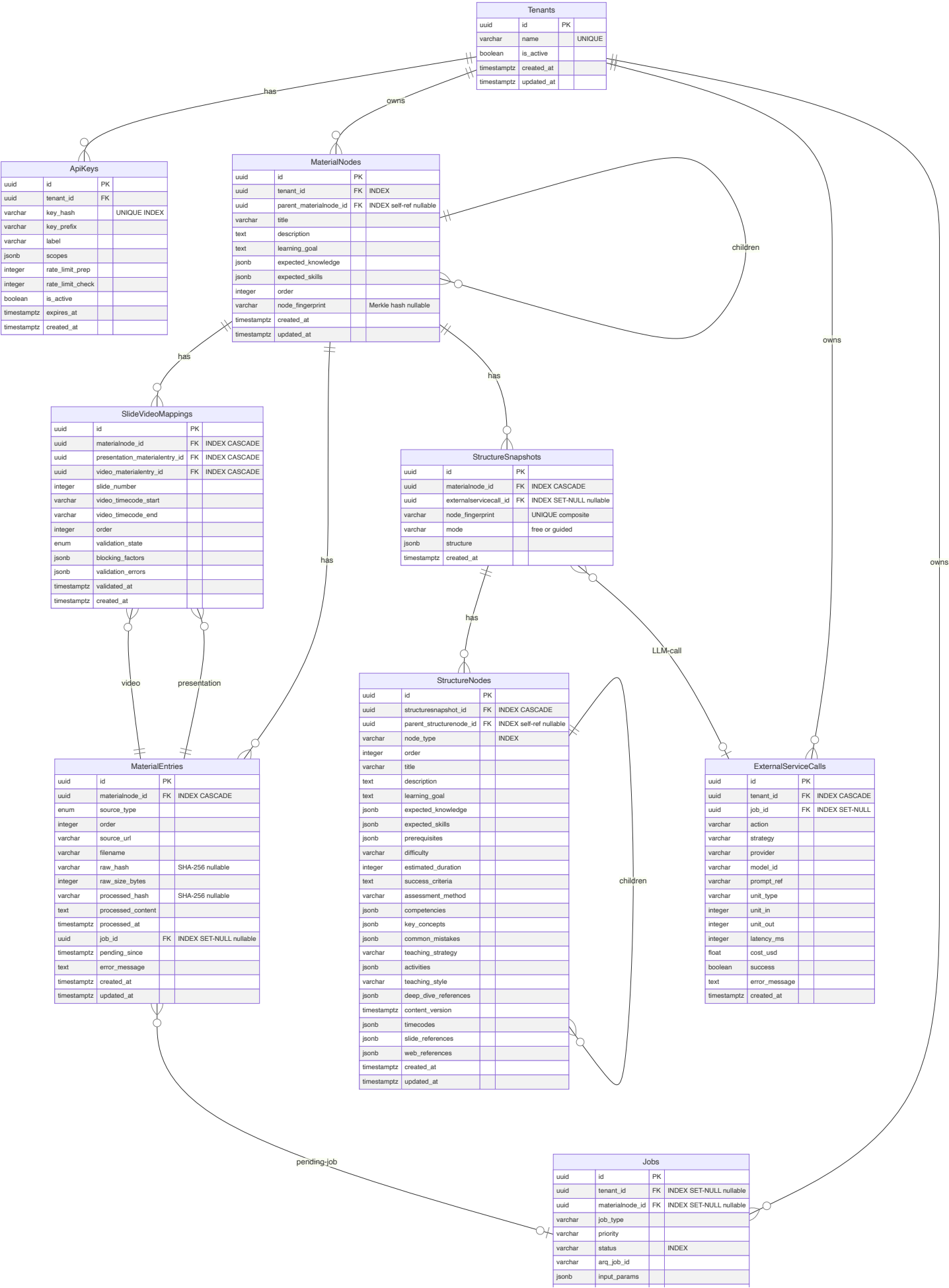
Database Schema — course-supporter

Auto-generated from ORM models (`src/course_supporter/storage/orm.py`).

Migration: up to `f6a7b8c9d0e1` (FK rename + DB comments).

All PKs are UUIDv7 (time-ordered). Timestamps use `TIMESTAMPZ` .

ER Diagram (Mermaid)



jsonb	depends_on		
text	error_message		
timestamptz	queued_at		
timestamptz	started_at		
timestamptz	completed_at		
timestamptz	estimated_at		

Table Descriptions

1. tenants

Comment: Multi-tenant organizations

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
name	VARCHAR(200)	NO	—	UNIQUE	
is_active	BOOLEAN	NO	true		
created_at	TIMESTAMPTZ	NO	now()		
updated_at	TIMESTAMPTZ	NO	now()	onupdate now()	

Indexes:

- tenants_pkey — PRIMARY KEY (id)
- ix_tenants_name — UNIQUE (name)

Relationships:

- tenants 1 → N api_keys (cascade: all, delete-orphan)

2. api_keys

Comment: Authentication keys with scope-based access control

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
tenant_id	UUID	NO	—	FK → tenants.id ON DELETE CASCADE	
key_hash	VARCHAR(64)	NO	—	UNIQUE , INDEX	SHA-256 hash of the API key. Raw key is never stored
key_prefix	VARCHAR(16)	NO	—		First 8 chars of the key for identification in logs
label	VARCHAR(100)	NO	'default'		
scopes	JSONB	NO	[]		JSON array of granted scopes (prep, check)
rate_limit_prep	INTEGER	NO	60		
rate_limit_check	INTEGER	NO	300		
is_active	BOOLEAN	NO	true		
expires_at	TIMESTAMPTZ	YES	—		
created_at	TIMESTAMPTZ	NO	now()		

Indexes:

- `api_keys_pkey` — PRIMARY KEY (`id`)
- `ix_api_keys_key_hash` — UNIQUE (`key_hash`)

Foreign Keys:

- `tenant_id` → `tenants.id` ON DELETE CASCADE

Relationships:

- `api_keys` N → 1 `tenants`

3. material_nodes

Comment: Hierarchical tree of course materials. Root node (parent IS NULL) = course

Column	Type	Nullable	Default	Constraints	Comment
<code>id</code>	UUID	NO	UUIDv7	PK	
<code>tenant_id</code>	UUID	NO	—	FK → <code>tenants.id</code> ON DELETE CASCADE, INDEX	
<code>parent_materialnode_id</code>	UUID	YES	—	FK → <code>material_nodes.id</code> ON DELETE CASCADE, INDEX	Self-referential FK. NULL = root node (course level)
<code>title</code>	VARCHAR(500)	NO	—		
<code>description</code>	TEXT	YES	—		
<code>learning_goal</code>	TEXT	YES	—		Pedagogical goal text for guided generation mode
<code>expected_knowledge</code>	JSONB	YES	—		
<code>expected_skills</code>	JSONB	YES	—		
<code>order</code>	INTEGER	NO	0		
<code>node_fingerprint</code>	VARCHAR(64)	YES	—		Merkle hash of content subtree. NULL = stale
<code>created_at</code>	TIMESTAMPZ	NO	<code>now()</code>		
<code>updated_at</code>	TIMESTAMPZ	NO	<code>now()</code>	<code>onupdate now()</code>	

Indexes:

- `material_nodes_pkey` — PRIMARY KEY (`id`)
- `ix_material_nodes_tenant_id` — (`tenant_id`)
- `ix_material_nodes_parent_materialnode_id` — (`parent_materialnode_id`)

Foreign Keys:

- `tenant_id` → `tenants.id` ON DELETE CASCADE
- `parent_materialnode_id` → `material_nodes.id` ON DELETE CASCADE (self-referential)

Relationships:

- `material_nodes` N → 1 `tenants`
- `material_nodes` N → 0..1 `material_nodes` (parent)
- `material_nodes` 1 → N `material_nodes` (children, cascade: all, delete-orphan)
- `material_nodes` 1 → N `material_entries` (cascade: all, delete-orphan)
- `material_nodes` 1 → N `slide_video_mappings` (cascade: all, delete-orphan)

- `material_nodes` 1 → N `structure_snapshots` (cascade: all, delete-orphan)

4. `material_entries`

Comment: Individual learning materials (video, presentation, text, web)

Column	Type	Nullable	Default	Constraints	Comment
<code>id</code>	UUID	NO	UUIDv7	PK	
<code>material_node_id</code>	UUID	NO	—	FK → <code>material_nodes.id</code> ON DELETE CASCADE, INDEX	FK to parent MaterialNode in the tree
<code>source_type</code>	ENUM('video', 'presentation', 'text', 'web')	NO	—	<code>source_type_enum</code>	
<code>order</code>	INTEGER	NO	0		
<code>source_url</code>	VARCHAR(2000)	NO	—		
<code>filename</code>	VARCHAR(500)	YES	—		
<code>raw_hash</code>	VARCHAR(64)	YES	—		SHA-256 of uploaded raw file for integrity detection
<code>raw_size_bytes</code>	INTEGER	YES	—		
<code>processed_hash</code>	VARCHAR(64)	YES	—		SHA-256 of processed_content for Merkle tree
<code>processed_content</code>	TEXT	YES	—		
<code>processed_at</code>	TIMESTAMPZ	YES	—		
<code>job_id</code>	UUID	YES	—	FK → <code>jobs.id</code> ON DELETE SET NULL, INDEX	FK to in-flight ingestion Job. NULL when idle
<code>pending_since</code>	TIMESTAMPZ	YES	—		
<code>error_message</code>	TEXT	YES	—		
<code>created_at</code>	TIMESTAMPZ	NO	<code>now()</code>		
<code>updated_at</code>	TIMESTAMPZ	NO	<code>now()</code>	<code>onupdate now()</code>	

Derived property (not in DB): `state` — one of `raw`, `pending`, `ready`, `integrity_broken`, `error` (computed from `error_message`, `job_id`, `processed_content`, hashes).

Indexes:

- `material_entries_pkey` — PRIMARY KEY (`id`)
- `ix_material_entries_materialnode_id` — (`materialnode_id`)
- `ix_material_entries_job_id` — (`job_id`)

Foreign Keys:

- `materialnode_id` → `material_nodes.id` ON DELETE CASCADE
- `job_id` → `jobs.id` ON DELETE SET NULL

Relationships:

- material_entries N → 1 material_nodes
- material_entries N → 0..1 jobs (pending job)

5. slide_video_mappings

Comment: Presentation slide to video timecode mappings

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
materialnode_id	UUID	NO	—	FK → material_nodes.id ON DELETE CASCADE, INDEX	FK to owning MaterialNode
presentation_materialentry_id	UUID	NO	—	FK → material_entries.id ON DELETE CASCADE, INDEX	
videomaterialentry_id	UUID	NO	—	FK → material_entries.id ON DELETE CASCADE, INDEX	
slide_number	INTEGER	NO	—		
videotimecode_start	VARCHAR(20)	NO	—		
videotimecode_end	VARCHAR(20)	YES	—		
order	INTEGER	NO	0		
validation_state	ENUM('validated', 'pending_validation', 'validation_failed')	NO	'pending_validation'	mapping_validation_state_enum	Enum: pending_validation, validated, validation_failed
blocking_factors	JSONB	YES	—		JSONB array of reasons preventing validation
validation_errors	JSONB	YES	—		
validated_at	TIMESTAMP TZ	YES	—		
created_at	TIMESTAMP TZ	NO	now()		

Indexes:

- slide_video_mappings_pkey — PRIMARY KEY (id)
- ix_svm_materialnode — (materialnode_id)
- ix_svm_pres_materialentry — (presentation_materialentry_id)
- ix_svm_video_materialentry — (video_materialentry_id)

Foreign Keys:

- materialnode_id → material_nodes.id ON DELETE CASCADE
- presentation_materialentry_id → material_entries.id ON DELETE CASCADE
- video_materialentry_id → material_entries.id ON DELETE CASCADE

Relationships:

- slide_video_mappings N → 1 material_nodes

- slide_video_mappings N → 1 material_entries (presentation)
- slide_video_mappings N → 1 material_entries (video)

6. structure_snapshots

Comment: LLM-generated course structure versions

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
materialnode_id	UUID	NO	—	FK → material_nodes.id ON DELETE CASCADE, INDEX	FK to target MaterialNode (root = whole course)
externalservicecall_id	UUID	YES	—	FK → external_service_calls.id ON DELETE SET NULL, INDEX	
node_fingerprint	VARCHAR(64)	NO	—	part of UNIQUE index	Merkle hash at generation time for idempotency
mode	VARCHAR(20)	NO	—	part of UNIQUE index	free or guided
structure	JSONB	NO	—		JSONB of generated course structure
created_at	TIMESTAMP	NO	now()		

Indexes:

- structure_snapshots_pkey — PRIMARY KEY (id)
- ix_snapshots_materialnode — (materialnode_id)
- ix_structure_snapshots_externalservicecall_id — (externalservicecall_id)
- uq_snapshots_identity — UNIQUE (materialnode_id , node_fingerprint , mode)

Foreign Keys:

- materialnode_id → material_nodes.id ON DELETE CASCADE
- externalservicecall_id → external_service_calls.id ON DELETE SET NULL

Relationships:

- structure_snapshots N → 1 material_nodes
- structure_snapshots N → 0..1 external_service_calls
- structure_snapshots 1 → N structure_nodes (cascade: all, delete-orphan)

7. structure_nodes

Comment: Recursive tree of generated course structure elements

Column	Type	Nullable	Default	Constraints	
id	UUID	NO	UUIDv7	PK	
structuresnapshot_id	UUID	NO	—	FK → structure_snapshots. id ON DELETE CASCADE, INDEX	
parent_structurenode_id	UUID	YES	—	FK → structure_nodes.id ON DELETE CASCADE, INDEX	Self-referential FK
node_type	VARCHAR(30)	NO	—	INDEX	Enum: module, lecture, ...

Column	Type	Nullable	Default	Constraints	
order	INTEGER	NO	0		
Section 1: Formal & organisational (Methodologist agent)					
title	VARCHAR(500)	NO	—		
description	TEXT	YES	—		
learning_goal	TEXT	YES	—		
expected_knowledge	JSONB	YES	—		
expected_skills	JSONB	YES	—		
prerequisites	JSONB	YES	—		
difficulty	VARCHAR(20)	YES	—		
estimated_duration	INTEGER	YES	—		
Section 2: Results & assessment					
success_criteria	TEXT	YES	—		
assessment_method	VARCHAR(50)	YES	—		
competencies	JSONB	YES	—		
Section 3: Methodological accents					
key_concepts	JSONB	YES	—		JSONB array of
common_mistakes	JSONB	YES	—		
teaching_strategy	VARCHAR(50)	YES	—		
activities	JSONB	YES	—		
Section 4: Context & adaptivity					
teaching_style	VARCHAR(50)	YES	—		
deep_dive_references	JSONB	YES	—		
content_version	TIMESTAMPZ	YES	—		
Section 5: Material references (Indexer agent)					
timecodes	JSONB	YES	—		JSONB of video
slide_references	JSONB	YES	—		

Column	Type	Nullable	Default	Constraints	
web_references	JSONB	YES	—		
Timestamps					
created_at	TIMESTAMPTZ	NO	now()		
updated_at	TIMESTAMPTZ	NO	now()	onupdate now()	

Indexes:

- structure_nodes_pkey — PRIMARY KEY (id)
- ix_structure_nodes_structuresnapshot_id — (structuresnapshot_id)
- ix_structure_nodes_parent_structurenode_id — (parent_structurenode_id)
- ix_structure_nodes_node_type — (node_type)

Foreign Keys:

- structuresnapshot_id → structure_snapshots.id ON DELETE CASCADE
- parent_structurenode_id → structure_nodes.id ON DELETE CASCADE (self-referential)

Relationships:

- structure_nodes N → 1 structure_snapshots
- structure_nodes N → 0..1 structure_nodes (parent)
- structure_nodes 1 → N structure_nodes (children, cascade: all, delete-orphan)

8. jobs

Comment: Background task queue entries (ingestion, generation)

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
tenant_id	UUID	YES	—	FK → tenants.id ON DELETE SET NULL, INDEX	
materialnode_id	UUID	YES	—	FK → material_nodes.id ON DELETE SET NULL, INDEX	FK to target MaterialNode. NULL for orphaned jobs
job_type	VARCHAR(50)	NO	—		
priority	VARCHAR(20)	NO	'normal'		
status	VARCHAR(20)	NO	'queued'	INDEX	
arq_job_id	VARCHAR(100)	YES	—		
inout_params	JSONB	YES	—		JSONB of task-specific parameters
depends_on	JSONB	YES	—		JSONB array of Job UUIDs that must complete first
error_message	TEXT	YES	—		
queued_at	TIMESTAMPTZ	NO	now()		

Column	Type	Nullable	Default	Constraints	Comment
started_at	TIMESTAMPTZ	YES	—		
completed_at	TIMESTAMPTZ	YES	—		
estimated_at	TIMESTAMPTZ	YES	—		

Indexes:

- jobs_pkey — PRIMARY KEY (id)
- ix_jobs_tenant_id — (tenant_id)
- ix_jobs_materialnode_id — (materialnode_id)
- ix_jobs_status — (status)

Foreign Keys:

- tenant_id → tenants.id ON DELETE SET NULL
- materialnode_id → material_nodes.id ON DELETE SET NULL

Relationships:

- jobs N → 0..1 tenants
- jobs N → 0..1 material_nodes
- jobs 1 → N material_entries (back-ref via job_id)

9. external_service_calls

Comment: Audit log of all external API calls (LLM, transcription, etc.)

Column	Type	Nullable	Default	Constraints	Comment
id	UUID	NO	UUIDv7	PK	
tenant_id	UUID	YES	—	FK → tenants.id ON DELETE CASCADE, INDEX	
job_id	UUID	YES	—	FK → jobs.id ON DELETE SET NULL, INDEX	
action	VARCHAR(100)	NO	''		
strategy	VARCHAR(50)	NO	'default'		
provider	VARCHAR(50)	NO	—		Service type: llm, transcription, embedding, etc.
model_id	VARCHAR(100)	NO	—		
prompt_ref	VARCHAR(50)	YES	—		
unit_type	VARCHAR(20)	YES	—		
unit_in	INTEGER	YES	—		JSONB of request payload sent to external service
unit_out	INTEGER	YES	—		JSONB of response received from external service
latency_ms	INTEGER	YES	—		
cost_usd	FLOAT	YES	—		

Column	Type	Nullable	Default	Constraints	Comment
success	BOOLEAN	NO	true		
error_message	TEXT	YES	—		
created_at	TIMESTAMPTZ	NO	now()		

Indexes:

- external_service_calls_pkey — PRIMARY KEY (id)
- ix_external_service_calls_tenant_id — (tenant_id)
- ix_external_service_calls_job_id — (job_id)

Foreign Keys:

- tenant_id → tenants.id ON DELETE CASCADE
- job_id → jobs.id ON DELETE SET NULL

Relationships:

- external_service_calls N → 0..1 tenants
- external_service_calls N → 0..1 jobs

Enums

Enum Name	Values
source_type_enum	video , presentation , text , web
mapping_validation_state_enum	validated , pending_validation , validation_failed

Application-level enums (stored as VARCHAR , validated in code):

Enum	Column	Values
GenerationMode	structure_snapshots.mode	free , guided
StructureNodeType	structure_nodes.node_type	module , lesson , concept , exercise
MaterialState	<i>(derived, not stored)</i>	raw , pending , ready , integrity_broken , error

FK Naming Convention

All foreign key columns follow the {tablename}_id pattern, where tablename is the referenced table's __tablename__ with underscores removed:

Referenced Table	FK Column Pattern	Example
tenants	tenant_id	api_keys.tenant_id
material_nodes	materialnode_id	material_entries.materialnode_id
material_entries	materialentry_id	slide_video_mappings.presentation_materialentry_id
structure_snapshots	structuresnapshot_id	structure_nodes.structuresnapshot_id
structure_nodes	structurenode_id	structure_nodes.parent_structurenode_id
external_service_calls	externalservicecall_id	structure_snapshots.externalservicecall_id
jobs	job_id	material_entries.job_id

Self-referential FKs use the `parent_` prefix:

- `material_nodes.parent_materialnode_id`
- `structure_nodes.parent_structurenode_id`